

# BRIAN JARRETT

## Senior Systems Engineer

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### PROFESSIONAL SUMMARY

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Highly analytical engineering professional with a 30-year foundation in rigorous system validation, data integrity, and autonomous technology adaptation. Proven track record of entering unfamiliar technical ecosystems, rapidly mastering complex architectures, and becoming a Subject Matter Expert (SME) for the organization's technology stack. Utilizes a strict practice of verifying data quality at the source to prevent the propagation of bad data and ensure the integrity of the final output, and architects robust ETL pipelines that merge disparate inputs into authoritative sources of truth. An objective, fact-based troubleshooter who translates intricate system data into actionable engineering and operational workflows, seeking to apply this rigorous systems engineering foundation to the pipeline architectures and infrastructure required to engineer and deploy AI/LLM workloads.

### TECHNICAL SKILLS

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#### Data & Systems Engineering:

- Python (Pandas, NumPy, Flask, SQLAlchemy)
- ETL Pipeline & Data Validation
- PostgreSQL, MySQL, Redis, MongoDB
- Systems/Robotics Design

#### Systems & Distributed Scale:

- Cisco UCS & Fabric Interconnect
- OpenStack (Nova, Heat)
- vSphere/ESXi, XenServer, Proxmox/KVM
- Ceph Block Storage (RBD)
- High-Availability (HA) Clustering

#### DevOps, Security & Leadership:

- Ansible, SaltStack
- CI/CD (GitHub Actions, Jenkins)
- FreeIPA/LDAP, RBAC/HBAC
- FedRAMP Compliance
- Qualys Vulnerability/Compliance
- Technical Leadership & Mentoring
- Advanced Diagnostics & Troubleshooting

### PROFESSIONAL EXPERIENCE

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#### Cisco | Cloud Engineering Technical Leader

**Aug 2019 - Present (Remote) \* Infrastructure Lead:** Lead operations of critical compute and storage systems for the Webex for Government (FedRAMP) cloud. Served as infrastructure team lead, managing system availability, platform uptime, and SLA tracking. \* **Data Platform Engineering:** Designed and operated a vulnerability management system using a custom ETL pipeline to extract data from **Jira API** and synchronize it to a **PostgreSQL** database. Built a **Python (Pandas/SQLAlchemy)** web frontend for service owners. Connected Tableau dashboards for executives to show current SLA metrics and the state of vulnerability remediation. \* **Compliance Automation:** Developed Python reporting tools extracting data from **STIG XCCDF (XML)** files and merging it with compliance data using the **Qualys API** to generate automated, audit-ready compliance checklists. \* **Technical Guidance:** Provided technical guidance for the Webex for Government environment, mentoring engineering teams in troubleshooting methodologies and collaborating on the resolution of complex vulnerability remediation issues to maintain service stability. \* **Skills:** Python (Pandas, SQLAlchemy), PostgreSQL, Jira/Qualys APIs, ETL Pipelines, Tableau, Ansible, RHEL/Ubuntu, Cisco UCS, STIG Compliance, FedRAMP.

#### Cisco Systems | Cloud Engineer - Webex

**Jan 2017 - Aug 2019 (Remote) \* Identity & Access Management:** Designed and deployed a **High-Availability (HA) FreeIPA cluster** using **Ansible** to manage RBAC/HBAC for **2,000+ servers, storage, and network infrastructure** across multiple data centers, replacing an aging, high-overhead **389-ds** environment. \* **Virtualization & Security:** Administered large-scale compute environments using **vSphere** on **Cisco UCS** hardware. Managed Qualys vulnerability scanners and orchestrated remediation efforts in commercial and FedRAMP sites, while maintaining core services such as bastion hosts and LDAP servers. \* **Skills:** Ansible,

FreeIPA, OpenStack, vSphere, Cisco UCS, RHEL/CentOS, Windows Server, RBAC/HBAC, Qualys, FedRAMP Compliance.

### Select Group @ Cisco | Cloud DevOps/QA Engineer (Contract)

**Feb 2016 - Jan 2017 (Remote) \* Performance Optimization:** Identified and removed redundant **Nova Scheduler** filters in **OpenStack**, reducing instance scheduling time for 700 nodes from 20 seconds to 5 seconds (a **75% improvement** in orchestration speed). \* **Cloud Infrastructure:** Configured and evaluated **RHUI (Red Hat Update Infrastructure)** and **Proxmox** clusters for InterCloud Services. Automated bare-metal server configuration using **Ansible** via lights-out management (LOM) interfaces. Troubleshoot and resolved a critical OpenStack **cloudbase-init** race condition on Windows images that was causing constant reboots. \* **Cloud QA:** Managed global test cycles for Compute and Heat components during cloud upgrades. Developed **Python, Bash**, and **Ansible** scripts to perform comprehensive automated orchestration tests. \* **Skills:** OpenStack, Ansible, Python, Bash, RHUI, Proxmox, CentOS/RHEL/Windows, Performance Tuning, Cloud QA.

### Australian Laboratory Services | Senior Systems Administrator

**Sep 2014 - Jan 2016 (Grand Junction, CO) \* Storage & Virtualization:** Engineered an Ubuntu-based **Ceph cluster** for **VMware ESXi** hosts, benchmarking iSCSI vs. RBD performance. Used **Ceph RBD snapshots** to manage automated schema testing, enabling a major production upgrade with zero errors. \* **Data Integrity & Backups:** Wrote custom **Python (SQLAlchemy)** verification scripts to check for missing foreign keys during migration, preventing data corruption. Configured backups via **LVM snapshots** and **rsync**. \* **Infrastructure & Automation:** Implemented **SaltStack** to manage Linux VMs and automate provisioning of Ceph nodes. Installed **Nagios** and **Observium** to monitor network operations and decrease service restarts. \* **Network Engineering:** Troubleshoot and resolved a complex routing issue where VoIP/Asterisk traffic was throttled by backup data tunnels; doubled bandwidth by replacing legacy gateway hardware. \* **Skills:** Ubuntu, Ceph RBD, VMware ESXi, SaltStack, Python, SQLAlchemy, Nagios, Observium, MySQL, OpenVPN, StrongSwan, Asterisk.

### Garfield County School District 16 | Senior Systems/Network Administrator

**Oct 2002 - Jun 2014 (Parachute, CO) \* Systems Integration & Automation:** Programmed a custom **Flask (SQLAlchemy)** app syncing Student Info Systems with **Active Directory** and **Google Apps**, reducing account helpdesk tickets by **97%**. \* **Network & VoIP Engineering:** Designed a high-speed WAN using **Ruckus wireless bridges**, increasing bandwidth by **6,500%** (1.5Mbps to 100+ Mbps) with zero monthly service costs. Deployed district-wide Cisco CallManager. \* **Compute Modernization:** Centralized desktop computing with **Windows Terminal Servers** and **HP Thin Clients** for 6 labs, saving **\$60k+** in CAPEX. Replaced legacy databases with centralized PostgreSQL platforms, and automated state compliance reporting via custom PHP scripts. \* **Skills:** Python, Flask, SQLAlchemy, PHP, PostgreSQL, XenServer, VirtualBox, Cisco Routers/Switches/PIX, Cisco CallManager, Ruckus Wireless.

### Foundational Experience

**1991 - 2002 \* Software QA Lead (StorageSoft/Phoenix Tech):** Managed testing and support departments. Structured QA testing workflows for **ImageCast** disk cloning tools and designed Windows/Linux test lab environments. \* **Airborne Radar Journeyman (U.S. Air Force):** Repaired high-power microwave systems, high-voltage circuits, and analog/digital systems on AWACS aircraft. \* **Skills:** Analog & Digital Diagnostics, VMS/PC Architecture, Disk Cloning (ImageCast), Software QA, Technical Training.

### EDUCATION & CERTIFICATIONS

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\* \*\*Regis University:\*\* C Programming/Unix courses \* \*\*Community College of the Air Force:\*\* Avionics/Electronics Technician - RADAR Specialist (AWACS) \* \*\*Colorado School of Mines:\*\* 2 Years (Electrical Engineering/Physics focus)